Gene Pulser® Electroprotocol

Cell Type Fungal / Yeast

Species Used *Schizosaccharomyces pombe*

Molecules Electroporated DNA: supercoiled plasmids, linear fragments for integration.

Before the Pulse

Cell Growth Medium YE or dropout media (recipes in paper)

Growth Phase at Harvest 1 x 10 (7) cells / ml

Pre-pulse Incubation none

Wash Solution 1.2 M sorbitol (ice cold, filter sterilized)

The Pulse

Instruments Used Gene Pulser® apparatus & Pulse Controller

Electroporation Temperature (Ice cold) 0 ° C

Electroporation Medium 1.2 M sorbitol

Cell Density 1 x 10 (4) cells / ml

Volume of Cells 200 µl

DNA Concentration 1 ng to 1 µg DNA per pulse

DNA Resuspension Buffer 1.2 M sorbitol

Volume of DNA <10 µl

Cuvette Gap 0.2 cm

Voltage 2.25 kV

Field Strength 1.125 kV/cm

Capacitor 25 µF

Resistor (Pulse Controller) 200 Ω

Time Constant 5 msec

After the Pulse

Outgrowth Medium SD + necessary nutrients

Outgrowth Temperature 30 °C

Length of Incubation 4 to 6 days

Selection Method or Assay Used auxotrophy

Electroporation Efficiency 1 x 10 (5) to 1 x 10 (6) for autonomous plasmids

Per Cent Survival ~50% (seeTable 1 in reference)

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Reference: Prentice, H. *Nucleic Acid Res*. **20** (3):621.

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176